

Following a Molecule Through the Nephron

Javalab Movement of Substances in the Nephron · Javalab

Senior Secondary (Years 11-12)

~30 minutes

Science · Biology

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://javalab.org/en/nephron_en/

Curriculum links: body systems · homeostasis

Learning intentions

- I can describe the three main processes (filtration, reabsorption, secretion) that occur in a nephron.
- I can track which substances are filtered, reabsorbed or secreted at each stage.
- I can explain how the nephron helps maintain a stable internal environment (homeostasis).

Getting started

Open the Javalab Movement of Substances in the Nephron simulation. Step through each stage of the nephron in order, starting with filtration at the glomerulus, and note which substances move at each stage before continuing.

Tasks

1. At the filtration stage, list the substances that pass out of the blood and into the nephron's filtrate.

2. Identify one substance that is NOT filtered out at this stage, and explain why (e.g. based on size).

3. Step through filtration, reabsorption and secretion. For each stage, record one example substance that moves and the direction it moves.

Stage	Example substance	Direction of movement

4. Explain why almost all of the glucose that is filtered out at the glomerulus needs to be reabsorbed back into the blood.

5. Identify one substance that is secreted directly from the blood into the nephron, rather than being filtered at the glomerulus. Why might the body need this extra pathway?

6. Using the stages you've stepped through, explain how the nephron's activity results in urine that is very different in composition from the original blood plasma.

7. Explain how the nephron helps maintain homeostasis, referring to at least one specific substance, such as water, salt or glucose, that the body needs to keep at a stable level.

Extension (optional)

Dialysis machines are designed to copy some of the jobs a nephron normally does. Using what you've learned about filtration and reabsorption, explain why dialysis alone cannot fully replace a working kidney's more selective reabsorption processes.
